
a Who is in each group
PAM on the % change in fibre area (mixed, type I, type II) and mCSA; gap selects K = 1, K = 2 used
Hi n = 8, Jaccard 0.89; Lo n = 8, Jaccard 0.88
PC2 (24%)
PC1 (74%)
Hi
Lo
HR
LR
b Proteins per contrast
1,900 tested; none at FDR < 0.05
Hi
Lo
Hi
Lo
T1
T2
T3
Training
Acute
proteins (down l up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
45 biopsies, 1,900 proteins
PERMANOVA
time R² 0.06, p = 0.0009
group R² 0.07, p = 0.42
PC2 (9.3%)
PC1 (13.9%)
Phenotype
31 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.4%)
PC1 (61.0%)
Hi
Lo
T1
T2
T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
Hi - Lo
Interaction
Striated muscle contraction
Regulation of muscle contraction
Citric acid cycle TCA cycle
Proteasome: CLEC7A dectin 1 signaling
Mitochondrial gene expression
Eukaryotic translation elongation
Striated muscle contraction
mRNA splicing
Protein DNA complex organization
Sarcomere organization
MYC targets v1
Cardiac muscle tissue development
Regulation of muscle contraction
Aerobic respiration and respiratory elect... *
Mitotic prometaphase *
Calcium signaling pathway *
Regulation of intracellular transport *
ATP synthesis coupled electron transport
Mitochondrial protein import
Mitochondrial membrane organization
Mitochondrial respiratory chain complex a...
Mitochondrial protein degradation
Citric acid cycle TCA cycle
Fatty acid beta oxidation
Mitotic spindle
Acyl CoA biosynthetic process
Cardiac muscle contraction
Glial cell differentiation
RHO GTPase effectors
Cytoskeleton dependent intracellular tran...
Complex iv assembly
Anchoring of the basal body to the plasma...
Mitochondrial biogenesis
Mitochondrial membrane organization
Cytoplasmic microtubule organization
Hi
Lo
Hi
Lo
T1
T2
T3
Training
Acute
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
up (n = 73), 82% same way, FDR 0.016
down (n = 29), 76% same way, FDR 0.026
Lo training sets in Hi's ranking
up (n = 59), 90% same way, FDR 0.14
down (n = 46), 74% same way, FDR 0.25
Hi training sets in Lo's ranking
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
up (n = 29), 76% same way, FDR 0.0088
down (n = 44), 86% same way, FDR 0.0014
Lo acute sets in Hi's ranking
up (n = 101), 57% same way, FDR 0.44
down (n = 88), 77% same way, FDR 0.42
Hi acute sets in Lo's ranking
rank in the other group's acute contrast, up to down (n = 1900